Hemang Negandhi

Senior Data Engineer | BFSI Data Engineering | BI & Analytics | AI-Driven Data Systems

Mumbai, India | Open to Remote / Hybrid Roles | +91 9321573159 | hemangcnegandhi@gmail.com

linkedin.com/in/hemang-negandhi | hemangnegandhi.github.io

PROFESSIONAL SUMMARY

BFSI-focused Data Engineering and Business Intelligence professional with 8+ years of experience in data analytics, ETL/ELT pipeline development, financial data processing, reconciliation automation, and AI-powered systems. Master of Science in Computer Science from the University of Texas at Arlington. Strong hands-on expertise in Python, SQL, Power BI (DAX), Tableau, and Alteryx Designer, with experience applying Machine Learning (ML) and Artificial Intelligence (AI) — building Generative AI and Agentic AI solutions using Large Language Models (LLMs), Prompt Engineering, Retrieval-Augmented Generation (RAG), and LangChain. Focused on enabling data-driven decision-making through robust data workflows, analytics, and data governance in the BFSI sector.

WORK HIGHLIGHTS

Project	What I Built	Impact
Financial Data Processing (BNP Paribas, C3 Project)	Process BFSI financial data monthly using Alteryx Designer, maintaining Data Governance and Data Quality standards.	**422 GB** processed monthly; BI insight for **10+** teams
Automated Financial Reconciliation (BNP Paribas, AXIOM–TRITON)	End-to-end reconciliation pipeline between AXIOM and TRITON using PowerShell, Alteryx, and Python; real-time regulatory reporting to Federal regulators via Power BI.	**900+ hours saved** annually
REST API Integration & Automation (AKA Enterprise Solution)	Dynamic analytics tool using the BambooHR REST API and Power BI; Data Quality improvements and HR Process Automation.	Adopted by **15 plus** stakeholders
ML Recommendation System (Master's Project, UT Arlington)	Scalable ML solution using Python, Flask, and GCP for predictive recommendations, applying NLP (stemming, lemmatization) and Feature Engineering.	**1.5M+** movie rows processed

RECENT AI WORK

Autonomous AI Job-Search Agent | Python, Groq API (LLaMA 3.3-70B), JobSpy, BeautifulSoup, Pydantic, Pandas, httpx

- Built an autonomous AI job-search agent in Python using JobSpy, REST APIs, and BeautifulSoup to scrape 50+ listings per run from LinkedIn, Indeed, and RemoteOK.
- Designed a multi-stage data pipeline with rule-based hard filtering, Pydantic validation, and URL-based deduplication — eliminating irrelevant results before LLM processing and reducing API token consumption by ~60%.
- Engineered an LLM scoring engine using the Groq API (LLaMA 3.3-70B) with structured JSON prompt engineering and explainable 1–10 relevance scoring against a dynamic candidate profile.
- Automated end-to-end data output using Pandas to generate dated, score-sorted CSV reports.
- Applied AI agent design patterns — tool-using agent architecture, ReAct-style reasoning, cost-optimized LLM orchestration, and modular separation of scraping, scoring, filtering, and output layers — using Python, dotenv, and httpx.

TECHNICAL SKILLS

Languages: Python, SQL, PowerShell
Databases: PostgreSQL, MS SQL (T-SQL), Oracle 11g (PL/SQL), MySQL
Data Engineering: Alteryx, ETL/ELT, Data Warehousing, Data Modeling (Star/Snowflake Schema), SSRS, Analytics
Business Intelligence: Power BI (DAX), Tableau, Qlik Sense, QlikView, SSIS
AI / ML: Machine Learning, Generative AI, LLMs (Groq/LLaMA 3.3-70B), Prompt Engineering, RAG, LangChain, Agentic AI
Frameworks / Libraries: Flask, Apache Spark (PySpark), Pandas, NumPy, NLP (NLTK), Pydantic
Cloud Platforms: AWS (Redshift, EC2, S3), Google Cloud Platform (GCP), Microsoft Azure
Web: HTML, PHP, JavaScript
Methodology: CI/CD, Data Governance
Tools / IDEs: PyCharm, Visual Studio Code

WORK EXPERIENCE

Senior Data Engineer | BNP Paribas, Mumbai, India **Jul 2020 – Present**

- Analyze and design Alteryx workflows for ETL; perform performance tuning and query optimization in Tableau.
- Create Tableau Data Extracts (TDEs) for multiple projects and schedule them on Tableau Server per business requirements.
- Coordinate with Paris and Madrid teams to deliver business requirements and ad hoc requests.
- Maintain and enhance C3FINANCE, a core data source processing 422 GB of data per month and delivering analytics to 10+ teams; improved data availability and reduced downstream query time by 30%.

- Lead migration of the financial data hierarchy from FIREF to GRECO_PMA, increasing data granularity from 12 levels to 30 levels and expanding insights by 25%.
- Spearhead the Deposit and Trend Analysis project, saving 786 work hours per year and reducing data errors from 10% to 2.5%; develop custom measures for financial forecasting and risk analysis, improving prediction accuracy by 25%.
- Engineer an automated reconciliation pipeline between AXIOM and TRITON reporting systems using PowerShell (scheduled file detection every 10 minutes), Alteryx macros, and embedded Python; match records using a composite key of Date + Series + Version.
- Save 900+ hours of manual work annually by automating the end-to-end reconciliation workflow with real-time email alerts at each stage and built-in error logging.
- Build a real-time regulatory reporting feed into Power BI from automated reconciliation outputs, enabling users to validate figures and submit accurate, timely reports to Federal regulators in the NAR region.
- Design version-aware, counter-based file detection logic to independently identify new or updated files on both drives, ensuring reconciliation triggers only on validated matched sets — improving accuracy by 75% and reducing errors from partial or duplicate file arrivals.

Business Data Analyst | BNY Mellon Pershing, New Jersey, USA **Sep 2019 – Mar 2020**

- Developed Alteryx workflows for data extraction, transformation, and loading (ETL).
- Used SQL to extract data and create ad hoc reports for visualization on Tableau Server.
- Conducted rigorous Tableau data validation by matching and verifying figures against the database, resulting in a 95% accuracy rate.
- Contributed to the development of Tableau workbooks for the EPO and Corporate Finance departments.
- Applied knowledge of star and snowflake dimensional schemas.
- Collaborated with cross-functional teams to gather business requirements and translate them into effective BI solutions, improving decision-making efficiency.

Power BI Intern | AKA Enterprise Solution, New York, USA **May 2018 – Jul 2018**

- Migrated an in-house application from a Power BI content workspace to an app workspace.
- Participated in creating dynamic visualizations to track employee performance.
- Established REST API integration with BambooHR to provide continuous access to real-time employee data; the tool was adopted by 15+ HR team members, improving efficiency and productivity.

Software Engineer | Infogain India Pvt Ltd, Mumbai, India **Aug 2015 – Jul 2017**

- Collected data in QlikSense from Oracle, CSV files, MS SQL, and MySQL sources.
- Created dynamic, interactive dashboards in Tableau, Power BI, and QlikSense.
- Analyzed underlying data for discrepancies, investigated errors, and implemented incremental loading.
- Performed ETL using SQL Server Integration Services (SSIS) and QlikView.
- Created a real-time streaming dashboard with real-time KPIs in Tableau.
- Documented ETL/ELT projects and workflows, including supporting documentation for troubleshooting.

Intern | CloudFronts Technologies, Mumbai, India **Jun 2015**

- Gained hands-on experience with SSIS, SSRS, and Power BI.
- Worked with Oracle, SQL, Twilio, and Zendesk connectors to integrate data with Power BI.

Intern | Algonation, Mumbai, India **Dec 2014 – Feb 2015**

- Gathered client requirements and created a MySQL data model for a healthcare project.
- Created a website using HTML5, CSS3, PHP, JavaScript, and the CodeIgniter framework.

EDUCATION

University of Texas at Arlington, USA — Master of Science in Computer Science, 2019
K. J. Somaiya College of Engineering, India — Bachelor of Information Technology, 2015

ACADEMIC PROJECTS (MASTER'S, UT ARLINGTON)

Movie Suggestion System

- Created a Flask application in Python 3.6 using NumPy, NLTK, and Pandas for word processing, including stemming and lemmatization.
- Processed a database of 1.5M rows to provide genre-based prediction results and deployed the application on Google Cloud Platform.

Online Advising Simulation

- Developed a client-server multi-threaded application for an online student-teacher advising system.
- Implemented message queuing using the producer-consumer design pattern.